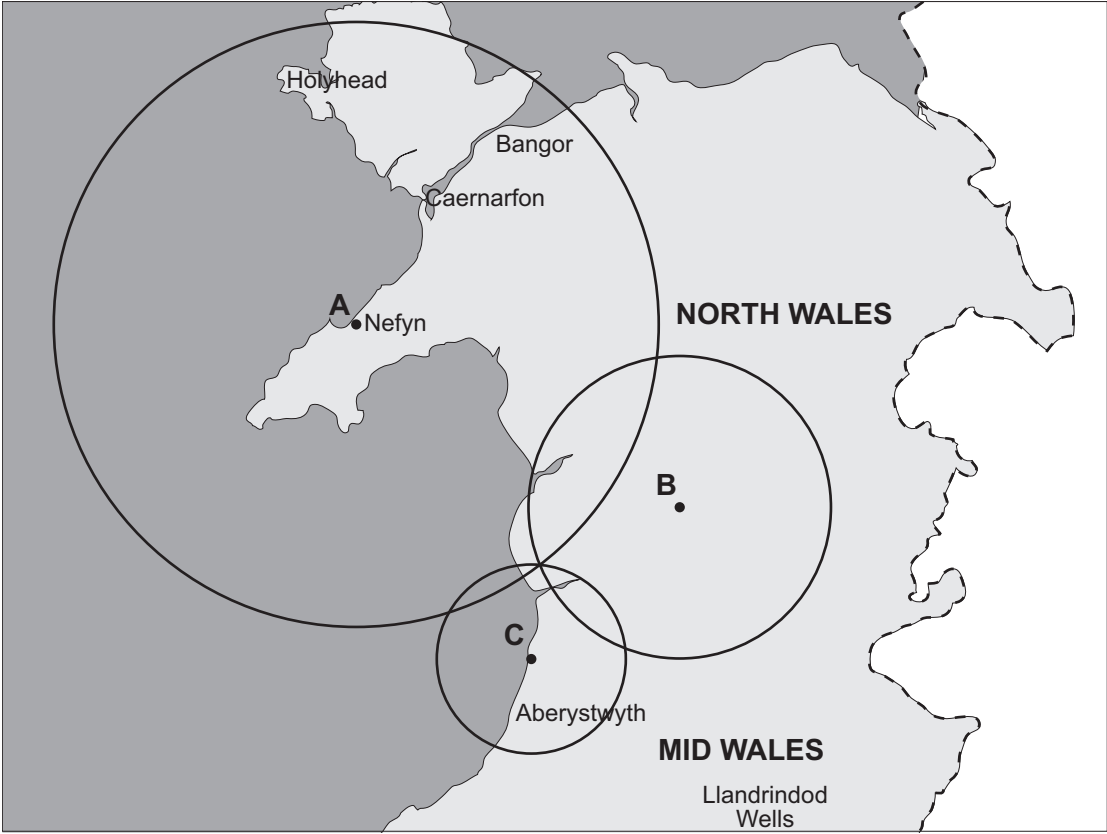


Answer all questions.

1. The waves from an earthquake can be monitored by seismological stations located on the surface of the Earth.



The diagram shows circles drawn around three seismological stations **A**, **B** and **C** in Mid and North Wales. The radius of each circle shows the distance of the epicentre from each station.

- (a) (i) **Label on the diagram** with a cross (X) the location of the epicentre. [1]

- (ii) Explain why at least three seismological stations are needed to locate the epicentre. [2]

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- (b) Explain which station (**A**, **B** or **C**) detected the earthquake first. [2]

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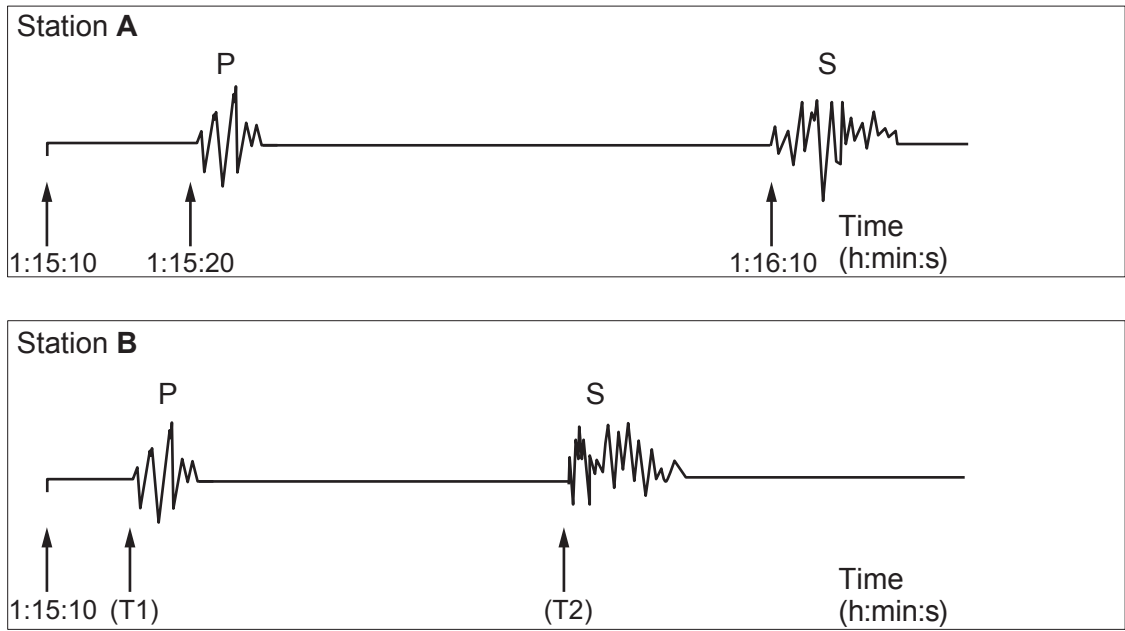
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- (c) The diameter of the circle around station A **is double** the diameter of the circle around station B.

The earthquake happened at 1 hour : 15 minutes : 10 seconds (1:15:10) in the morning.

The following signals were obtained at stations **A** and **B**.



Due to an error on the equipment the **arrival times** (T1 and T2) for the two signals were not recorded at station **B**.

Using signal information from station **A**, determine the times that the P waves and the S waves arrived at station **B**. Explain how you arrived at your answers. [3]

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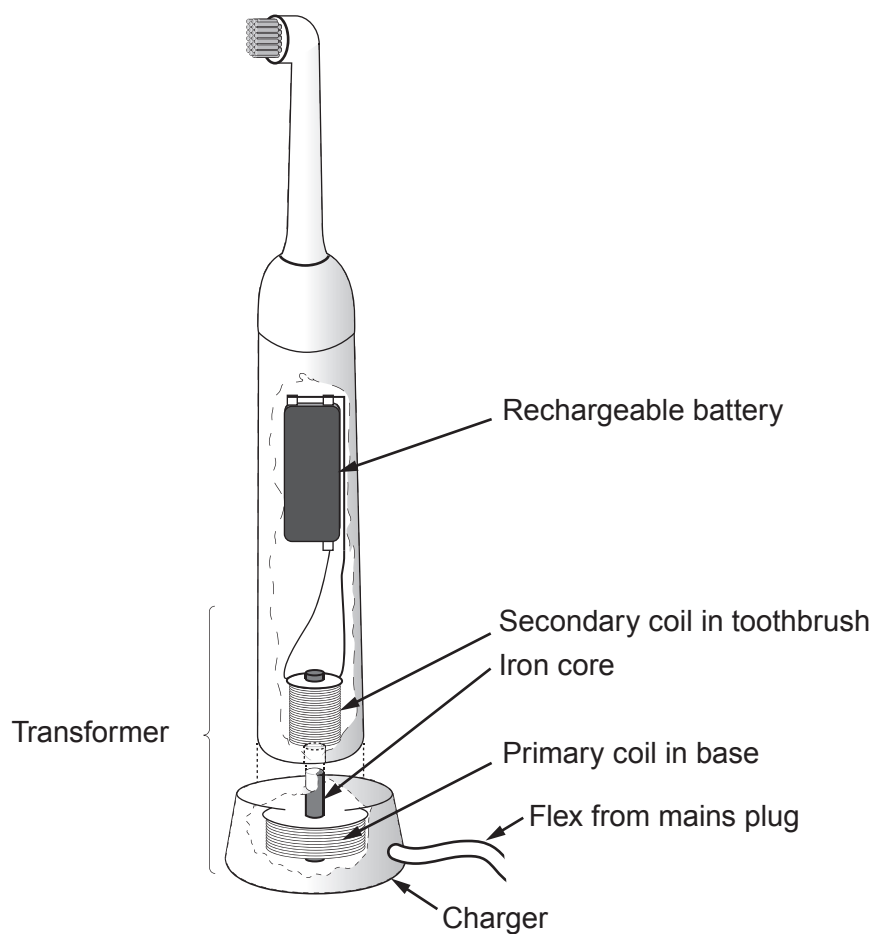
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Arrival time of P waves (T1) = ..... hours ..... minutes ..... seconds.

Arrival time of S waves (T2) = ..... hours ..... minutes ..... seconds.

3. The diagram shows a rechargeable toothbrush which contains a transformer.



(a) Explain how a voltage is produced in the secondary coil of the transformer. [3]

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(b) For the following questions, assume that the transformer is 100% efficient.

(i) Describe what is meant by *100% efficient*. [2]

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- (ii) The secondary coil inside the toothbrush has 480 turns. The primary coil in the base has 9 200 turns and is connected to the 230 V mains. Use an equation from page 2 to calculate the voltage across the secondary coil. [2]

Voltage = ..... V

- (iii) Given that the current produced in the secondary coil is 100 mA, use an equation from page 2 to calculate the power of the toothbrush. [3]

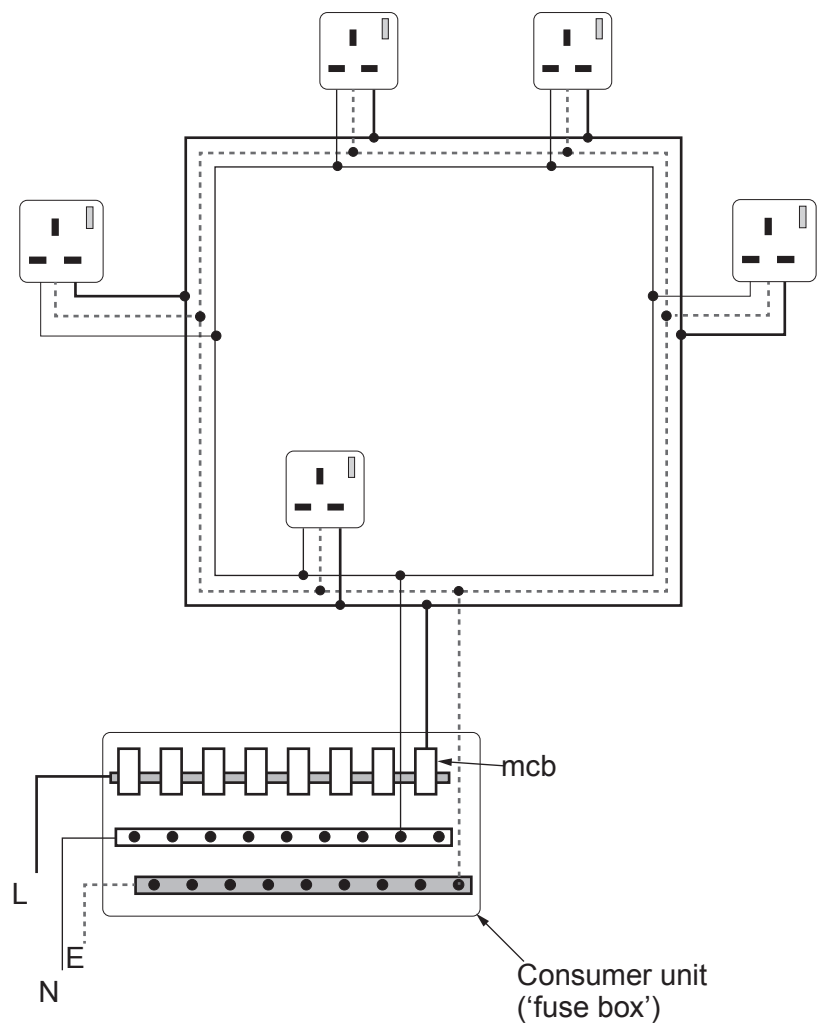
Power = ..... W

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4. (a) The diagram shows a simple ring main circuit for the kitchen of a house. The sockets are used to supply a tumble dryer with a current of 9 A, a kettle with 10 A and a toaster with 5 A all at the same time.



- (i) Describe the function of the earth wire. [2]

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- (ii) State **two** advantages of a miniature circuit breaker (mcb) over a fuse. [1]

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2. The **epicentre** is the point on the Earth’s surface directly above an earthquake. Seismic stations detect earthquakes by the tracings made on seismographs.

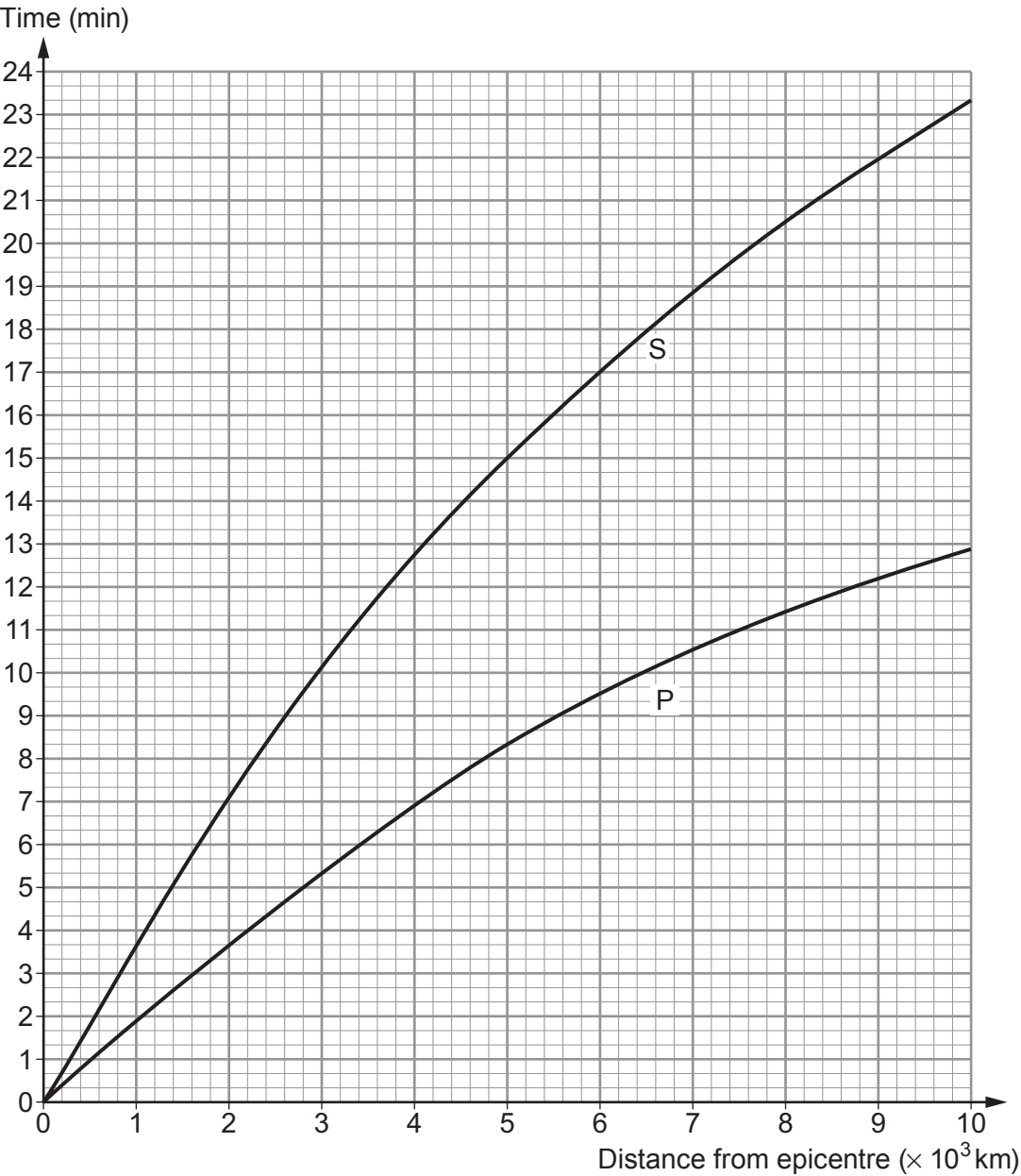
(a) Surface, P and S waves are three types of earthquake waves.  
Tick (✓) the boxes next to the **three** correct statements about earthquake waves. [3]

- Surface waves travel the fastest ☐
- S waves travel on the surface of the Earth ☐
- S waves are transverse waves ☐
- P waves travel through solids and liquids ☐
- P waves are longitudinal waves ☐
- S waves cause the most damage ☐

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(b) The graph shows the time taken by P and S waves to travel different distances from the epicentre.



Each small square on the time axis represents 20 s.

(i) Use the graph to answer the following questions.

I. State the time it takes for a P wave to travel  $5 \times 10^3$  km from the epicentre. [1]

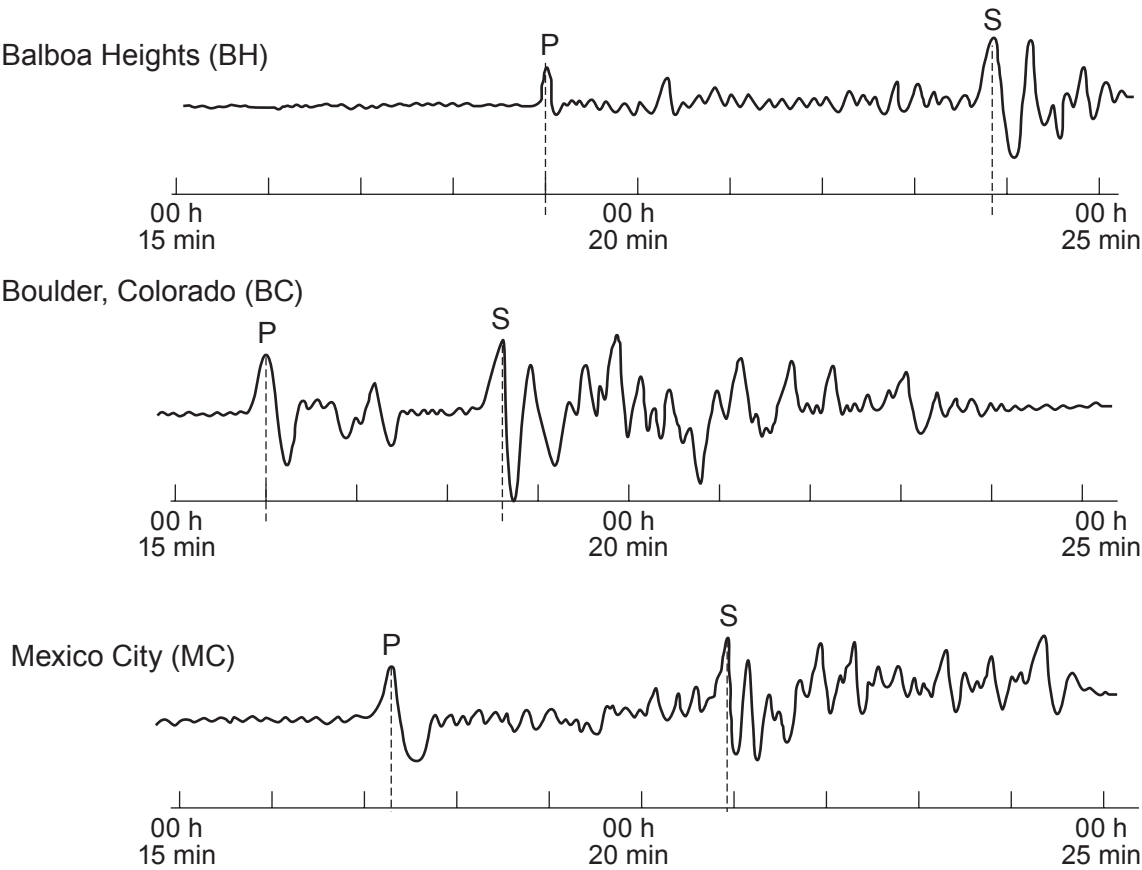
time = ..... min ..... s

II. State the **extra** time it takes S waves to travel  $5 \times 10^3$  km from the epicentre. [1]

time = ..... min ..... s



(ii) Study the three seismograph tracings below. Tracings made at three separate seismic stations are needed to locate an earthquake epicentre. P shows the arrival of P waves and S shows the arrival of S waves.

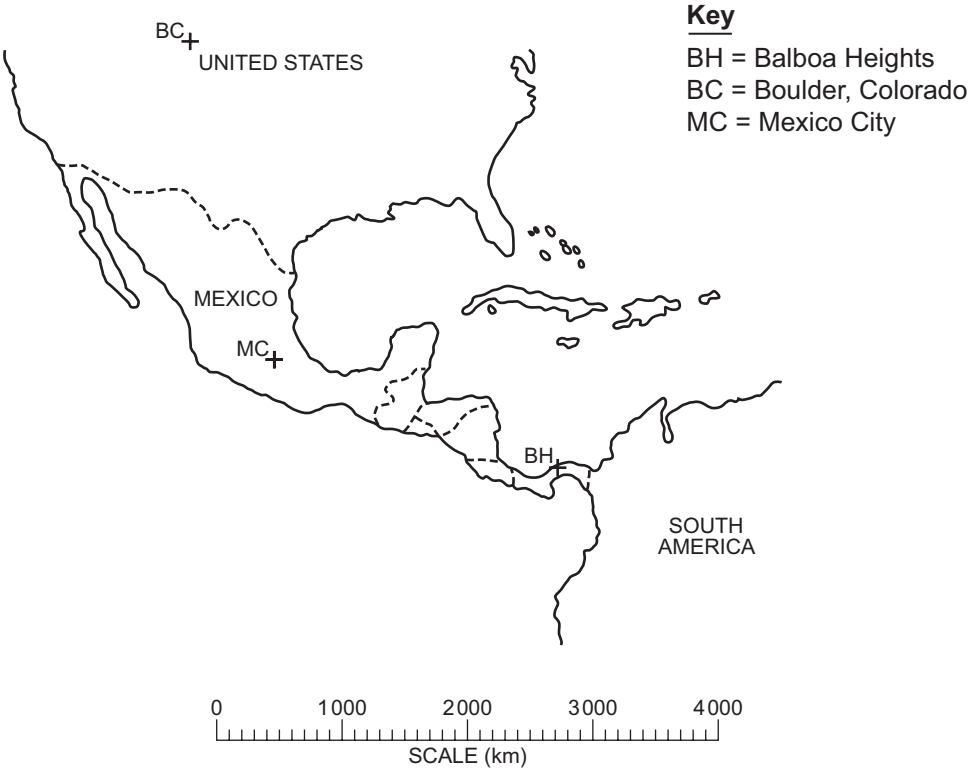


Use the information in the graph and tracings to **complete the table**. [3]

| City                   | Arrival time of P waves (h:min:s) | Arrival time of S waves (h:min:s) | Time difference for P and S waves (h:min:s) | Distance to epicentre ( $\times 10^3$ km) |
|------------------------|-----------------------------------|-----------------------------------|---------------------------------------------|-------------------------------------------|
| Balboa Heights (BH)    | 00:19:00                          | 00:23:50                          | 00:04:50                                    | 3.2                                       |
| Boulder, Colorado (BC) | ..... : ..... : .....             | 00:18:40                          | ..... : ..... : .....                       | .....                                     |
| Mexico City (MC)       | 00:17:15                          | 00:20:55                          | 00:03:40                                    | 2.2                                       |



(iii) Use the data in the table to find the epicentre of the earthquake on the diagram below. [3]



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5. Satellites are used to communicate between base stations at different places on Earth.

- (a) (i) **Complete the diagram** below to show how base station A communicates with base station B. [3]



- (ii) Explain why a geostationary satellite must be used for constant communication rather than a geosynchronous one. [2]

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- (b) A base station communicates with another base station using a single geostationary satellite  $3.6 \times 10^4$  km above the Earth. They communicate with each other using microwaves of wavelength 2.8 cm travelling at  $3 \times 10^8$  m/s.

Use an equation from page 2 to calculate the time delay between sending a signal from one base station and receiving the signal at the other. [4]

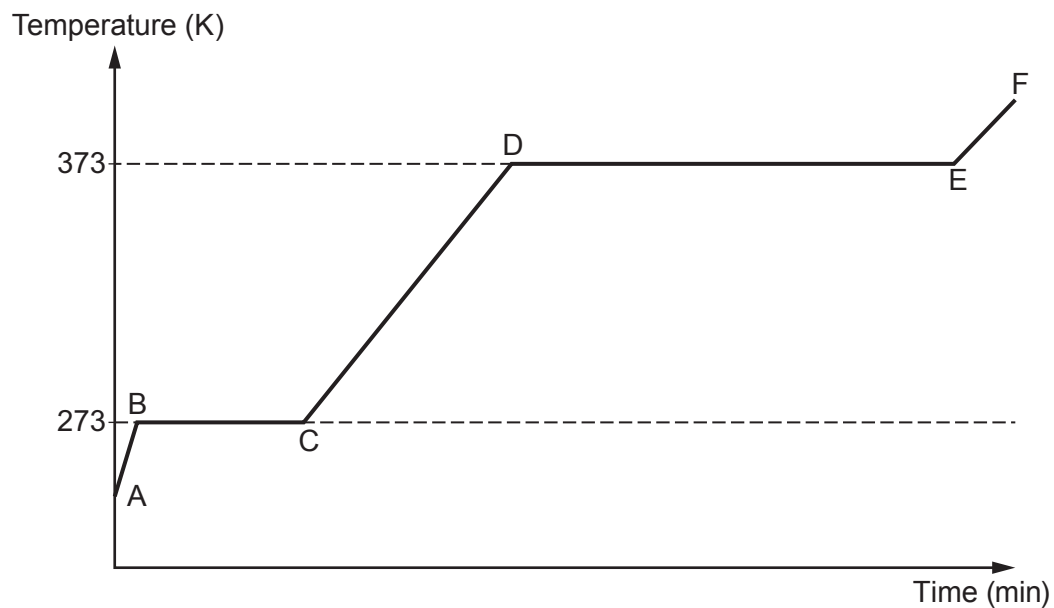
Time = ..... s

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6. The graph below shows how the temperature of a block of ice changes over time as it is heated.



- (a) Refer to the graph to explain the changes in temperature and changes in state of the block of ice in terms of the behaviour of molecules. [6 QER]



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(b) The specific latent heat,  $L$ , of fusion of ice is 336 000 J/kg.

(i) Explain what is meant by this statement. [2]

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(ii) The ice block has a mass of 800 g. Its initial temperature is 250 K.  
Use equations from page 2 to calculate the energy required to raise its temperature to 273 K and completely melt the ice block at 273 K.  
The specific heat capacity,  $c$ , of ice is 2030 J/kg K. [5]

Energy = ..... J

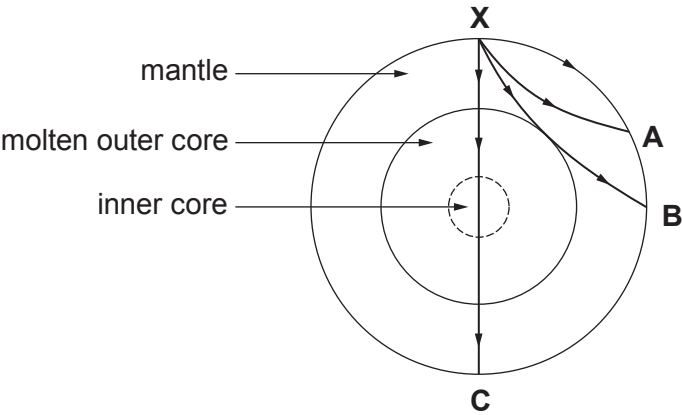
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6. Earthquake monitoring stations can detect P and S waves from earthquakes. A seismic trace contains information about times of arrival for each wave type.

(a) The diagram shows some paths taken by seismic waves travelling from an earthquake at **X**. There are three seismic monitoring stations: **A**, **B** and **C**.



Compare the seismic traces obtained at **A**, **B** and **C**, explaining any similarities **and** any differences. [6 QER]

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(b) S waves are transverse waves and P waves are longitudinal waves. Describe the difference between transverse and longitudinal waves.

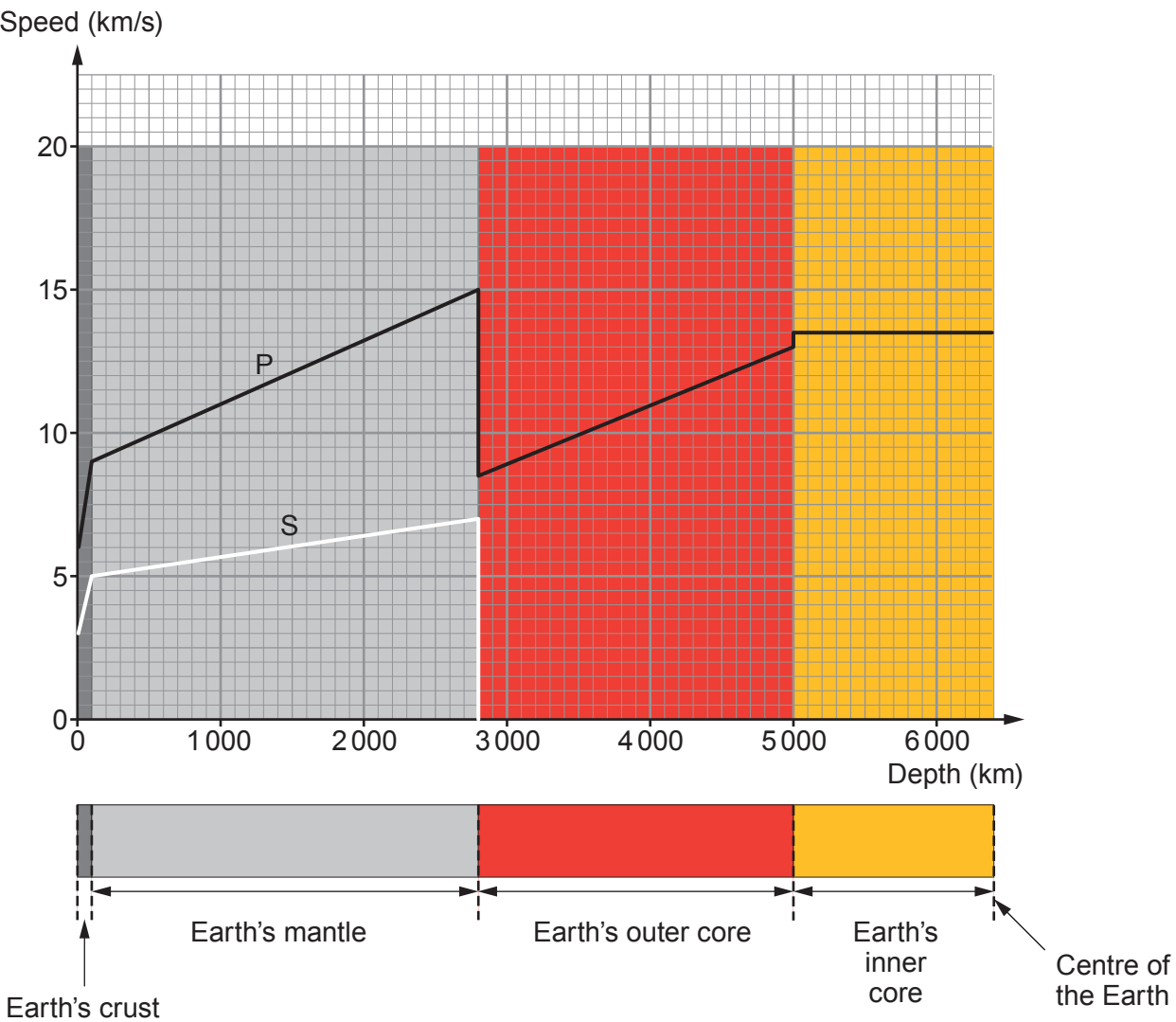
[2]

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2. P and S waves are produced during an earthquake. They travel at different speeds. The graph shows how the speed of P and S waves change with depth below the Earth's surface.



(a) P waves are longitudinal waves.  
Explain what is meant by a longitudinal wave. [2]

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(b) Use the graph to calculate the difference in speed between the P wave and the S wave at a depth of **100 km**. [2]

Speed difference = ..... km/s



- (c) (i) Bob claims that the maximum speed of P waves is greater in solids than in liquids. Explain, using data from the graph, whether Bob's claim is correct. [2]

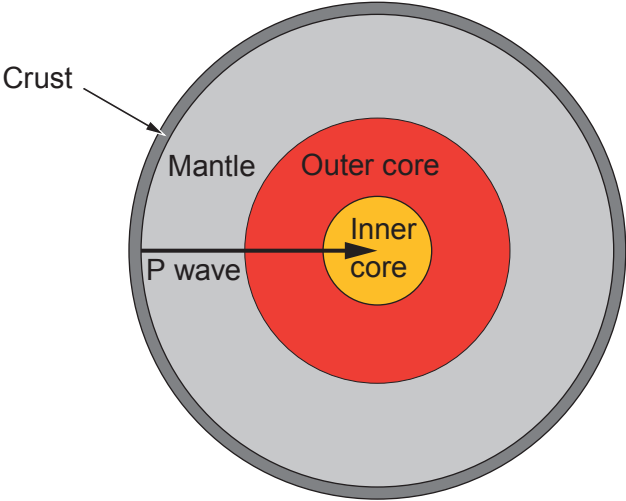
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- (ii) A P wave (shown in Diagram 1) travels a distance of 6 300 km in a straight line to the centre of the Earth in a time of 550 s.

Diagram 1



Bob states that the mean speed of the P wave over this distance is equal to its actual speed only at the depth of 3 500 km. Use an equation from page 2 to show whether Bob's statement is correct. [3]  
Space for calculations.

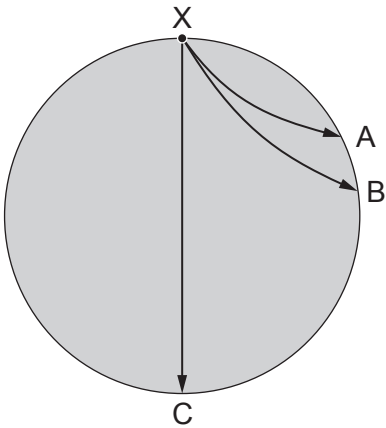
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(d) Diagram 2 shows waves passing through the Earth following an earthquake at point X.

Diagram 2



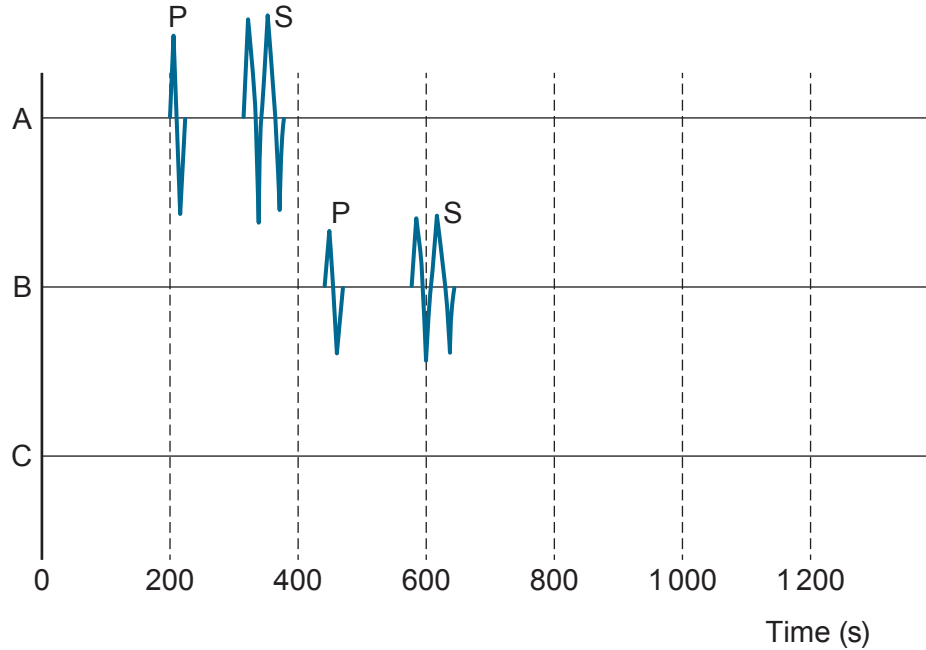
Monitoring stations at A, B and C detect the earthquake by signals that they receive.

- (i) Use the time given in part (c)(ii) to state the time taken for the P wave to travel from **X to C** (ignore the Earth's crust in this). [1]

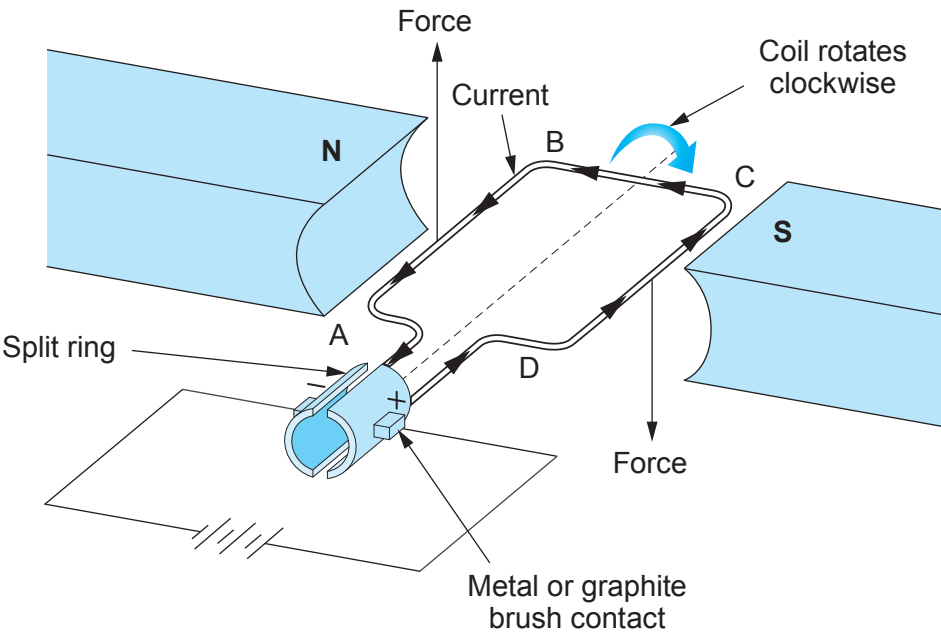
Time = ..... s

- (ii) The signals received at monitoring stations A and B are shown in Diagram 3. **Add to Diagram 3** the signal that you would expect to be received at station C. [3]

Diagram 3



7. The diagram shows a simple d.c. electric motor consisting of a single turn of wire ABCD.



(a) Describe how Fleming's left-hand rule can be used to show that the side **AB** of the wire moves upwards. [3]

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(b) By considering the current in **AB**, explain how the split ring allows the coil to continue to rotate. [2]

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- (c) The sides AB and CD have lengths of 6.0 cm and the side BC has a length of 3.0 cm. The magnetic field has a strength of 0.04 T and a current of 0.7 A passes through the coil.

(i) Use the equation:

force on a conductor (at right angles to a magnetic field) carrying a current = magnetic field strength × current × length

to calculate the force on side **AB** of the coil.

[3]

Force = ..... N

- (ii) Describe the effect on the force, on side **AB** of the coil, of doubling **both** the battery voltage **and** the number of coils.

[2]

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5. The table below gives examples of transverse and longitudinal waves.

| Transverse waves | Longitudinal waves |
|------------------|--------------------|
| light            | sound              |
| S waves          | P waves            |

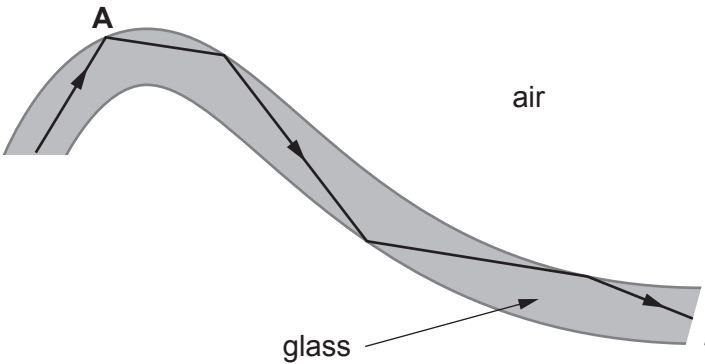
(a) Describe the difference between transverse and longitudinal waves. [2]

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(b) Light waves can travel through an optical fibre in the way shown in the diagram.



(i) Name the effect where light changes direction as shown at point **A**. [1]

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(ii) Explain why light waves travel through the fibre as shown. [2]

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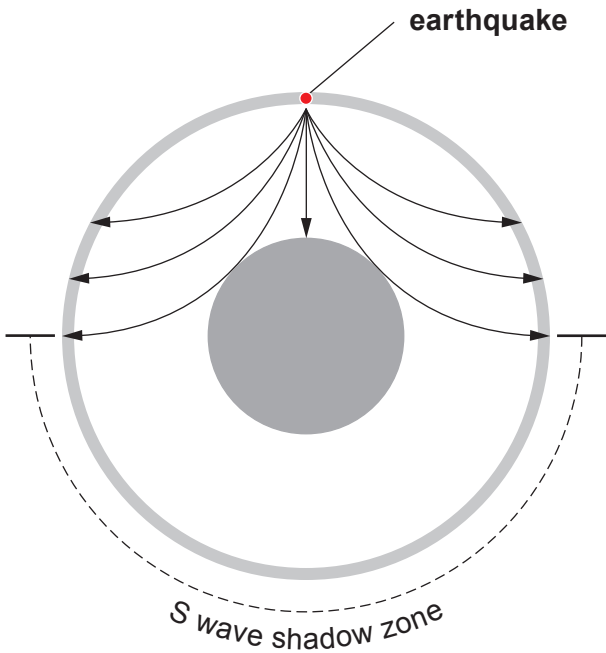
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(c) The diagram below shows an S wave shadow zone.



Explain how this led to the development of a model for the structure of the Earth. [2]

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